The Topology of Transformation: From Admissibility to Observability in Discrete Configuration Spaces

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Introduction: The Architecture of Transformational Continuation

The formal representation of dynamic systems within theoretical physics, applied mathematics, and algebraic topology has historically prioritized a static ontology. In classical paradigms, systems are defined primarily by their instantaneous states, and their chronological progression is governed by external algorithmic rules, continuous differential laws, or fixed combinatorial relations. This conventional approach implicitly treats the state as the fundamental entity and the transformation as a secondary, applied operation. Recent advancements in algebraic topology, particularly concerning discrete configuration spaces, graph braid groups, and combinatorial Hodge theory, provide the mathematical scaffolding to reverse this epistemological hierarchy. In this revised paradigm, a combinatorial or physical system is defined fundamentally by its space of admissible transformations, rendering the instantaneous state as a mere temporal cross-section of an ongoing relational transition.

This conceptual inversion forms the basis of the Nexus framework—a mathematical architecture wherein distinction, continuation, configuration, residual structure, and observability manifest as successive topological consequences of how a transformation system is represented, invariantized, and probed. Rather than proceeding from the hypothesis that a given system is a collection of distinct objects moving through an arbitrary background manifold, the architecture proceeds from two abstract primitives: the existence of a distinction, and the absolute requirement that every distinction must admit a lawful continuation.

When formalized, the existing mathematics of graph configuration spaces provides a highly controlled, nontrivial laboratory in which the geometric consequences of transformational continuation can be proved

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exactly. The analysis demonstrates that structural persistence, physical loss, and observational uncertainty are not arbitrary artifacts or statistical noise. Instead, they are precise geometrical manifestations of topological constraints placed upon a continuous transformation field. The objective is to establish a rigorous mathematical architecture capable of outlining what transformation inherently requires, proving that representations, invariant structures, relational geometries, and observer access occupy fundamentally distinct mathematical strata.

Following a strict methodological progression from proved mathematics to demonstrated model behavior, and finally to physical interpretation, this report reorganizes the theoretical findings around a central spine of topological progression: Admissibility, Representation, Invariance, Residual, Relation, and Access. Within this spine, the decoupling of the invariant 1-skeleton (transformation channels) from the invariant 2-complex (relations among channels) emerges as a central theorem. Furthermore, the integration of exact-rank matrix certification provides a proof-grade methodology for distinguishing dimensionally forced harmonic survivors from rank-certified vanishing classes, thereby refining the cohomological interpretation of topological memory and discrete curvature.

1. Admissibility: Configuration Spaces, Collision Exclusion, and Substrate Changes

To rigorously test the consequences of transformational continuation, one requires a mathematical arena where interacting transformations under defined exclusion constraints yield a computable topological space. Graph configuration spaces supply this precise machinery.

Given a finite, connected simple graph $X = (V, E)$, the unlabelled configuration space of K points on X, denoted $\mathrm{UConf}_K(X)$, is defined as the space of K-element subsets of the graph. The fundamental group of this space is the graph braid group, an algebraic object that encodes all possible collision-free exchanges of particles across the network. This continuous configuration space can be reduced to a discrete cellular or cubical complex, originally formulated via the Świątkowski complex or the Abrams model. The Świątkowski framework provides an established chain model for configuration-space homology, defined structurally as $S(\Gamma) = \mathbb{Z}[E] \otimes \otimes_{v \in V} S(v)$, which An, Drummond-Cole, and Knudsen explicitly recovered in their broader theory of subdivisional spaces. By viewing the graph and its subdivisions as a subdivisional space—a diagram object in a category of cell complexes—they proved that the homology of the configuration spaces is computed by the derived tensor product of the Morse complexes of the pieces of the decomposition.

The computational tractability of these spaces relies heavily on the discrete Morse theory developed by Farley and Sabalka. In discrete Morse theory, a discrete gradient vector field is constructed over the cell complex, inducing a classification of the cells into three mutually exclusive types: collapsible, redundant, and critical. The fundamental rule of this discrete gradient field dictates that for any p-cell $\alpha^{(p)}$, there can be at most one $(p+1)$-cell containing $\alpha^{(p)}$ for which the discrete Morse function evaluates lower, and at most one $(p-1)$-cell contained in $\alpha^{(p)}$ for which the function evaluates higher. By Forman's theorem, a CW complex endowed with such a discrete gradient vector field is homotopy equivalent to a streamlined complex comprising exactly m_p cells of dimension p, where m_p represents the number of critical cells of that dimension.

Farley and Sabalka applied this gradient field to the unlabelled discrete configuration spaces of graphs to provide a clear description of the critical cells, yielding algorithmic presentations for the braid groups of any tree. In this application, the generators of the braid group correspond precisely to the critical 1-cells of the configuration space, while the relators correspond to the critical 2-cells.

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When a physical or combinatorial system undergoes a change in its substrate $X \to X'$, the admissible transformation field $\mathcal{T}_X$ maps to $\mathcal{T}_{X'}$. Topological tracking of this mapping reveals exactly what it means for a system to change without requiring the absolute destruction and recreation of its history. Rigorous chain-map analysis of substrate transformations proves that structural dimension reduction in a configuration space is not synonymous with the annihilation of relational data. The change decomposes into precise modification mechanisms, which must be evaluated through a rigorous structural progression.

Definition: A change of substrate is a mapping $X \to X'$ that modifies the underlying vertices, edges, or capacity constraints of the transformation field, generating a relative restriction map Δ_R, a dual pullback F_R^*, or projection maps ρ_R and q_R^*.

Lemma: Structural modifications isolate into distinct algebraic operations. Extension ($\mathcal{T}_X \subset \mathcal{T}_{X'}$) yields a surjective relative restriction map Δ_R onto the residual with a zero cokernel. Re-factorization ($T \mapsto T_2 T_1$), such as edge subdivision, yields a surjective dual pullback F_R^*. Identification ($S_1 \sim S_2$) merges configurations, yielding $\ker q_R^* = 0$. Exclusion ($S \notin S_{\text{admissible}}$) deletes a configuration, forcing a downward map that yields $\ker \rho_R > 0$.

Theorem: Under the definitions of the mathematical architecture, geometric identification is topologically safe ($\ker q_R^* = 0$), while collision exclusion is strictly destructive ($\ker \rho_R > 0$). Change of representation does not equal the loss of an admissible continuation.

Computation: This dynamic is definitively proved via a "Capacity-2 Transport" control experiment, which computationally separates the act of identifying nodes from the act of forbidding states. Under a standard graph contraction $X' \to X$, the mathematical sequence factors into two mappings. The first map, $\rho: X' \to X_0'$, removes configurations containing both endpoints of the edge being merged, executing the exclusion boundary. The second map, $q: X_0' \to X$, performs the geometric identification. By relaxing the exclusion constraint at the merged site to permit a token capacity of 2, the collision states are no longer deleted. In all evaluated configuration space topologies (e.g., $C_7 \to C_6$, $Q_3' \to Q_3$), the square commutes exactly at the cochain level. The pullback map acts as a strict isomorphism between the exclusion kernel downstairs ($\ker \sigma$) and the collision kernel upstairs ($\ker \rho$).

Interpretation: The geometric contraction contributes no additional kernel; the defect is exclusively the relative contribution of the excluded star. Loss does not occur merely because two variables are merged into one. Loss occurs exclusively because the prior coexistence of states becomes inadmissible, severing the continuation graph and forcing transformations into a null space. Destruction is defined solely as the removal of a previously admissible continuation.

2. Representation: Coarse Arenas and the Spurious Coincidence

A principal discovery generated by overlaying the Nexus architecture onto graph configuration spaces is the strict mathematical separation of four ontological layers: the underlying configuration space, the chosen representation, the residual cohomological structure, and the observer attempting to read it. In empirical physics, these layers are frequently conflated, resulting in the assumption that an observer's inability to detect change implies the cessation of underlying physical transformations. Experiments within discrete configuration spaces force these layers apart, demonstrating precisely how structural artifacts and observational blindness manifest.

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Definition: A coarse representation maps an unrefined model into the true subdivisional configuration space via a pushforward or pullback mechanism, denoted structurally as **$\phi_*: H^1_{\rm coarse} \to H^1_{\rm true}$**.

Lemma: It is mathematically permissible for this mapping to possess a substantial kernel. The elements of this kernel are spurious representation classes—artifactual invariants manufactured by the coarse arena that simply do not exist in the true subdivisional space.

Theorem: Representation fidelity is mathematically distinct from measurement resolution. The presence of a correct numerical count of topological classes does not guarantee the existence of a correct underlying structure (correct count $\neq$ correct structure).

Computation: The danger of conflating representation fidelity with correct structural counting is vividly illustrated by the Betti number generation at the $K = 3$ and $K = 4$ token capacity thresholds. In specific chiral graphs, the coarse, un-subdivided arena manufactures higher-body artifactual classes that numerically match the birth count of genuine higher-body classes in the true configuration space. Both the coarse and true spaces produce an apparent expansion of classes, yielding an identical numerical increase. However, deeper relational tracking proves that this numerical agreement is an algebraic coincidence driven by entirely disparate geometric mechanisms.

Structural Feature	True Configuration Space	Coarse Arena
Origin of Expansion	Tail junctions (dead-end leaf storage)	Base junctions (single-edge branches)
Capacity Trajectory	Frozen at base, grows linearly at tail	Frozen at tail, traps tokens at base
Nature of Classes	Genuine higher-body relational interactions	Spurious single-occupancy artifacts

The true configuration space generates legitimate new distinctions because tail junctions store tokens, pushing the relational depth outward as capacity increases. Conversely, the coarse arena generates fake classes because its unrefined base junctions act as traps. This creates apparent token distributions that instantly vanish under a sufficiently fine subdivision.

Interpretation: This disproves the common heuristic that coarse representations are merely undersampled versions of reality akin to the Nyquist–Shannon sampling limit. Instead, a coarse arena actively fabricates false structures. The observable Betti count is not the ontological structure. Observable dimension is not enough to identify mechanism, validating the tenet that distinctions become operational only when they survive the constraints and resolutions of the system attempting to represent them.

3. Invariance: Symmetry Quotients and Parity Survival

The next topological layer governs how symmetrical observation alters the accessible transformation field. Expanding the symmetry group of an observer does not inherently destroy the underlying transformations.

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Definition: For a configuration space observed through an expanded symmetry group ($G \subset G'$), the invariant observation sector is modeled by the quotient graph and the associated pullback map $R_{G'} \hookrightarrow R_G$.

Lemma: If a symmetry operation $g \in G$ reverses an oriented edge orbit, any invariant antisymmetric 1-cochain ω must satisfy $\omega(e) = \omega(ge) = \omega(-e) = -\omega(e)$, algebraically forcing $\omega(e) = 0$. These flippable edge orbits mathematically collapse into topological self-loops in the quotient representation, contributing zero invariant degrees of freedom.

Theorem (The Cone-Quotient Theorem): Because the deletion of self-loops cannot alter the connected components of a graph, the remaining transformation structure forms the symmetry quotient graph Q_r. The dimension of the invariant residual sector, D_r, corresponds exactly to the first Betti number of this quotient graph:

$$\dim D_r = \dim H_G^1(Y, Y_0) = \beta_1(Q_r) = E_Q - V_Q + c_Q$$

Computation: The inclusion of the component term c_Q is mathematically essential. In fully saturated scenarios—where the capacity parameter reaches $K = |V| + r$—the newly admitted layer consists of a single configuration containing absolutely no vacancies. Without a vacancy, there is no cross transformation capable of reaching the relative cone point. Consequently, the quotient graph fragments into two isolated components ($c_Q = 2$), ensuring the Betti calculation perfectly accounts for the disconnected topology and returns $\dim D_r = 0$ instead of an invalid negative Betti number. At the terminal layer, $D_{K-1} = \#(N(m)/\mathrm{Stab}(m)) - 1$. The terminal residual vanishes ($D_{K-1} = 0$) if and only if the stabilizer group is fully transitive on the neighbors of m.

Classical geometric assumptions frequently dictate that overarching global structural symmetry unconditionally destroys flat topological memory, effectively stripping the system of all orientation legacy. However, algebraic mapping into quotient spaces definitively restricts this assertion.

Interpretation: The terminal zero observed in various capacity ladders is not evidence that a generalized thermodynamic price has been exhausted; it is a strict geometric consequence of local symmetry blinding the invariant observer. Global saturation and local symmetry constitute two vastly distinct mechanisms. Furthermore, flat invariant survival is highly contingent upon the parity of the symmetry operation.

When a physical substrate undergoes a mirrored geometric collapse, it aligns with the homological projection map $C^1 \to C^1/\mathrm{im}d_0$. Applying an orientation-reversing symmetry universally forces the fundamental homology class $[\omega] \to -[\omega]$, algebraically nullifying singular 1-token loops. Yet, multi-token flat classes can be distinctly reflection-even, successfully surviving the symmetric collapse. This divergence is exposed when comparing orientation-preserving (Chiral) and orientation-reversing (Mirror) symmetries:

Gadget Structure	**Symmetry Type**	**Token Parameter (K)**	**Total Residual (ϵ)**	**Exact Formula (Curved + Flat)**
Chiral Z_3 Base	Orientation-Preserving	K=2	29	$\epsilon = 4 + 25$

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Gadget Structure	Symmetry Type	Token Parameter (K)	Total Residual (ϵ)	Exact Formula (Curved + Flat)
Chiral Z_3 Base	Orientation-Preserving	K=3	150	$\epsilon = 5 + 145$
Mirror D_3 Base	Orientation-Reversing	K=2	18	$\epsilon = 1 + 17$
Mirror D_3 Base	Orientation-Reversing	K=3	148	$\epsilon = 1 + 147$

As explicitly observed in the mirror D_3 test, non-zero flat invariants (specifically, exactly 1 flat class at both $K = 2$ and $K = 3$) cleanly survive despite the overarching spatial orientation reversal. The absolute survival of a one-dimensional class within the deep kernel of the induced action of the graph automorphism explicitly denotes an algebraic state that is completely invariant under the destructive mirror operation. Thus, symmetry can hide structure without annihilating the underlying space, proving that relational memory survives parity collapse within the mathematical model.

4. Residual: Exact Rank Certification and Proof-Grade Methodology

Moving beyond existence proofs, the computational verification of these topological spaces relies on rigorous matrix certification. The cochain complex evaluating the invariant sector yields boundary matrices M_i mapping between $C^0 \to C^1 \to C^2$. The discrete differential operator inherently satisfies $d_1 d_0 = 0$.

Definition: The evaluation of the boundary matrices M_i over a finite field $\mathbb{F}_p$ via modulo-p reduction provides a rigid algebraic boundary for the matrix rank.

Lemma: Reduction modulo a prime p cannot increase the mathematical rank of a matrix over the rationals; it can only maintain it or decrease it due to modular collisions. Therefore, the following inequality holds strictly:

$$\mathrm{rank}_{\mathbb{F}_p} M_i \leq \mathrm{rank}_{\mathbb{Q}} M_i$$

Theorem: By computing the ranks over $\mathbb{F}_p$, one establishes an absolute lower bound on the image dimension, which directly translates to an upper bound on the kernel dimension. If the sum of the modulo-p ranks of M_0 and M_1 perfectly equals the total dimension of the 1-cochain space t, then the rational rank must also sum to t:

$$\mathrm{rank}_{\mathbb{F}_p} M_0 + \mathrm{rank}_{\mathbb{F}_p} M_1 = t \implies \mathrm{rank}_{\mathbb{Q}} M_0 + \mathrm{rank}_{\mathbb{Q}} M_1 = t$$

The harmonic sector, defined cohomologically by the quotient $H^1_G = \ker d_1 / \mathrm{im} d_0$, is therefore mathematically forced to vanish identically, guaranteeing $\epsilon_{\rm harm} = 0$.

Computation: This exact-rank result acts as an elegant, proof-grade certification theorem. Across an exhaustive computational evaluation of configuration spaces, this modular reduction algorithm was

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successfully applied to 102 independently certified cases, testing across two distinct cell predicates and two prime characteristics.

Interpretation: This robust computational methodology allows the framework to formally distinguish between spaces that are dimensionally forced to be nonzero and spaces that possess a rank-certified zero. It eliminates computational noise or floating-point ambiguity when evaluating the existence of harmonic memory, cementing the zero-side of the topological boundary with absolute rigor.

5. Relation: Harmonic Structure and The Separation of Complex Layers

The most critical conceptual consequence of the advanced configuration space mapping is the definitive mathematical separation of the 1-skeleton from the 2-complex. Previous paradigms frequently assumed that the topological complexity of a class intrinsically dictates its harmonic behavior. However, precise tracking of the differential map $C^1 \xrightarrow{d_1} C^2$ proves that the existence of harmonic structures relies on a strict internal hierarchy separating available transformation channels from the admitted relations among those channels.

Definition: The harmonic memory of a discrete configuration space is structurally governed by the invariant 1-cochains (C_G^1) and the 2-cell relations (C_G^2) applied to them.

Lemma: The dimension of the harmonic sector is governed by the rank-nullity theorem applied to the discrete differential operator acting on the invariant 1-cochains:

$$\epsilon_{\rm harm} = \epsilon - \operatorname{rank}(d_1\vert{}_{C_G^1})$$

Because the rank of the linear map d_1 cannot possibly exceed the dimension of its codomain C_G^2, this algebraic truth directly yields a rigorous existence obstruction:

$$\epsilon_{\rm harm} \ge \epsilon - \dim C_G^2$$

Theorem: Harmonic survival is not merely something discovered experimentally in localized graphs; it is governed by an absolute dimension obstruction. The obstruction theorem dictates that:

$$\epsilon > \dim C_G^2 \Longrightarrow H_G^1 \neq 0$$

Computation: The data confirms that this obstruction executes substantial geometric work. Across 222 exhaustive six-vertex configurations, 85 cases were dimensionally active. Of those 85 active cases, exactly 85 exhibited **$\epsilon_{\rm harm} > 0$**. Furthermore, 61 of the 85 cases were exactly tight, and 85 of the 98 total observed nonzero cases across all topologies were directly certified by this dimension bound.

Interpretation: This establishes a definitive new hierarchy for determining harmonicity:

$$\text{1-skeleton} \rightarrow \text{dimension obstruction} \rightarrow \text{boundary rank} \rightarrow H_G^1$$

Level 1 evaluates the dimension obstruction. If $\epsilon - \dim C_G^2 > 0$, the existence of a nonzero harmonic survivor is proven outright. Level 2 evaluates the rank; if the dimension obstruction is silent ($\epsilon \leq \dim C_G^2$), the outcome depends exclusively on the linear rank of the boundary map d_1. Level 3 requires evaluating detailed relation geometry when dimensions permit either outcome, as the actual incidence structure of the 2-cells ultimately dictates the rank.

This structural hierarchy is flawlessly demonstrated by evaluating the paw, bull, and wheel6 graph architectures:

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Graph Architecture	ϵ (C1 dim)	dimCG2	Obstruction (ϵ−dimCG2)	Harmonic Survivor (ϵharm)
Paw	0	0	0	0
Bull	2	1	1	1
Wheel6	3	3	0	0

The bull graph provides the cleanest demonstration of the obstruction theorem: possessing 2 residual channels and only 1 invariant relation, it is algebraically forced to possess at least one survivor, and it generates exactly one. The wheel6 graph illustrates that the obstruction is sufficient for existence but not strictly necessary for vanishing; a silent obstruction ($3 - 3 = 0$) does not automatically prove vanishing, though in the wheel6 topology, vanishing does indeed occur.

5.1 Predicate Dependence and the Separation of C^1 and C^2

The most profound realization of this hierarchy is predicate dependence. For an identical physical substrate X, token capacity K, and symmetry group G, the invariant 1-skeleton C_G^1 remains mathematically unchanged regardless of the 2-cell filling rules. However, altering the 2-cell predicate explicitly alters C_G^2, the boundary map d_1, and ultimately the harmonic residual H_G^1.

This was verified via an eight-vertex adversarial graph construction. Under both a disjoint-support square predicate and an all-4-cycle predicate, the 1-skeleton dimension remained locked at $\epsilon = 8$. However, under the disjoint-support squares construction, the system generated $\epsilon_{\rm harm} = 1$, whereas under the all-4-cycles construction, the harmonic survivor vanished entirely, yielding $\epsilon_{\rm harm} = 0$.

This adversarial divergence proves a vital topological distinction: ϵ is exclusively a property of the invariant 1-skeleton, while $\epsilon_{\rm harm}$ is exclusively a property of the invariant 2-complex. It cleanly maps the distinction between available transformation channels (C^1) and the admitted relations among those channels (C^2). The latter dictates which of the former actually close to form verifiable macroscopic memory.

5.2 Refinement of Prior Empirical Generalizations (Negative Results)

Consequently, previous empirical generalizations regarding harmonic vanishing must be formally corrected. Early data suggested a vanishing law where networks possessing two or fewer vertex orbits ($\#(V/G) \leq 2$) unconditionally guaranteed harmonic vanishing ($H_G^1 = 0$).

This statement is now definitively dead as a general mathematical law. The 8-vertex adversarial counterexample features exactly two vertex orbits yet generates $\epsilon_{\rm harm} = 1$ under the disjoint-support-square complex. Interestingly, the all-4-cycle construction maintained the vanishing rule ($112/112 = 0$).

Therefore, the correct scientific statement is not that "two vertex orbits cause harmonic vanishing." Rather, vertex-orbit count was an effective empirical predictor for specific complex generation rules, but its apparent vanishing law is fundamentally predicate-dependent, and it is verifiably false for the disjoint-

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support-square complex. Harmonic vanishing cannot be reduced to a single graph-theoretic invariant; it relies dynamically on the geometric incidence of the specific relations admitted by the system.

6. Access: Spectator Dependence and the Operator Spectrum

With the invariant relational structure established, one must evaluate how finite observer access dictates the resolution of the topological field. The complexity of the memory within a given configuration space is quantified by mapping the residual actions of the graph braid groups. Previous literature frequently applied group-theoretic Johnson filtrations to evaluate this memory depth. However, this terminology is potentially misleading in the current context. The derived machinery requires a much sharper representation decomposition: a Boolean-subset, Johnson-scheme representation decomposition analyzing spectator dependence.

Definition: Topological interactions occurring strictly within spectator subsets—where tokens entirely avoid the active moving tokens—can be evaluated by mapping the configurations onto the Johnson scheme $J(v,k)$. The interactions generate specific transfer maps T_p.

Lemma: The actual derived operator structure relies on scaled transposition matrices T_p defined exactly by:

$$T_p = \frac{1}{\binom{K-1}{p}} M^{\top}$$

with the subsequent operator $L_p = A_p T_p$. This maps the localized spectator interactions directly via the Johnson-scheme spectrum $\lambda_i(n,K,p)$.

Theorem: This operator structure yields a rigorous hierarchy of transfer and access statements. A continuous transformation model inherently contains specific thresholds for visibility. Specifically, the topological reach of a localized reader is bounded strictly by the environmental capacity:

$$\mathrm{reach}_p \leq \min\{\epsilon(X,p+1), \epsilon(X,K)\}$$

In the invertible transfer regime, this boundary achieves exact equality: $\mathrm{reach}_p = \epsilon(X,p+1)$. Furthermore, under the stated configuration-space hypotheses, the exchange theorem proves that the kernel of the exchange holonomy sits strictly within the zero-level residual space: $\ker(\operatorname{Hol}_{\rm exch}) = L_0$.

Computation: The evaluation of Eulerian transition systems provides the analytical bridge connecting macroscopic observations of partition readers to microscopic combinatorial structures. For a uniform random permutation on $N = 128$ cells, representing a total phase space of roughly 716 bits, a maximally coarse 1-cut reader recovers a mere 3.55 bits. The expected mutual information proves that the bits seen plus the bits hidden sum exactly to the total combinatorial entropy $\log_2 N!$.

Interpretation: The prior hypothesis that "higher-order distinctions require higher-order access" acts as the interpretive compression of these exact transfer theorems. A localized 1-body or 2-body reader scanning a topologically complex space will perceive a perfectly flat void, completely blind to higher-order relational topologies. Observer blindness is a consequence of spatial configuration, not the absence of physical transformation.

6.1 Cohomological Obstruction and Discrete Order Residue

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To quantify topological memory without overreaching into unverified cosmological claims, one relies on combinatorial Hodge theory and the discrete Hodge Laplacian. Historically, memory and internal state have been conceptually conflated with the intrinsic curvature of the underlying spatial manifold—a geometric dent in physical reality.

When multiple tokens inhabit a graph, their disjoint movements generate 2-cells (faces), structuring the configuration space as a specialized square complex. A differential cochain ω mapped onto this cubical complex quantifies the structural memory of the path. The non-commutative circulation of ω around a specific 2-cell represents the discrete order residue.

The magnitude and algebraic behavior of this commutator partition the invariant cochains into three rigorous limits: exact (perfectly agnostic to path history), flat (zero order residue on localized 2-cells but non-zero holonomy globally), and curved (non-zero circulation on discrete localized 2-cells).

What is mathematically established by the computations is a discrete cohomological classification mapping $C^1 \to C^2$ containing closed/non-closed cochains and harmonic residuals. The designation of the resulting quantity as "curvature" is exclusively a Nexus interpretation. The safer, rigorous epistemological hierarchy is:

cohomological obstruction → discrete order residue → Nexus interpretation as curvature

The mathematics does not prove that this is physical curvature in the differential-geometric or relativistic sense. Rather, it demonstrates that the modeled memory structure possesses a discrete cohomological realization that perfectly replicates the behavior of memory without requiring the introduction of a curved ambient manifold.

Conclusion

The mathematical architecture developed herein establishes that a system defined by distinguishable states and lawful continuations generates an uncompromising topological hierarchy. Through the integration of the Świątkowski subdivisional spaces, Farley-Sabalka discrete Morse theory, and combinatorial Hodge Laplacian mechanics, the configuration of transformation is cleanly parsed into successive strata: Admissibility, Representation, Invariance, Residual, Relation, and Access.

The deepest common thread uniting these independent computations is the rigorous definition of structural survival. Across all tests, the identical mathematical pattern emerges:

structure exists ≠ structure survives representation ≠ structure survives symmetry ≠ structure is observable

The most profound realization of this framework is the strict algebraic decoupling of transformation channels from the relations governing them, mapped precisely to the boundary differential $C^1 \xrightarrow{d_1} C^2$. Because $\epsilon_{\rm harm} \ge \epsilon - \dim C_G^2$, harmonic memory is an explicit dimensional consequence of the geometry of permitted relations, not an arbitrary anomaly. The loss of memory is not caused by symmetry or substrate identification, which safely quotient the space, but strictly by exclusion boundaries that sever the continuity of the transformation field.

Ultimately, this analysis proves that observable dimension is insufficient to identify the underlying mechanism, and that the inability of an observer to detect change is a function of the reader's own spatial lattice, not the cessation of kinetic reality. A transformation model can contain significantly more admissible structure than a given representation or restricted observer can actively access. Proposing that the physical

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universe mirrors this infinite density of hidden transformation acts as an interpretive physical hypothesis, built solidly upon the proved mathematics of discrete topological continuation.

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